Early Allergen Introduction and Baby-Led Weaning

Randomized trials show that introducing peanut protein between 4 and 11 months reduces peanut allergy by ~80% in high-risk infants (LEAP), while modified baby-led weaning and traditional spoon-feeding produce similar growth and choking rates (BLISS).

GRADE: HIGH | STATUS: DRAFT | DOMAIN: FEEDING, HEALTH | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

High-certainty randomized trial evidence (the LEAP trial) shows that introducing age-appropriate peanut protein between 4 and 11 months of age reduces the prevalence of peanut allergy at age five by approximately 81% compared with strict avoidance. Separate randomized evidence on complementary feeding method (the BLISS trial) shows that modified baby-led weaning—where infants self-feed soft finger foods while parents avoid high-risk choking hazards and include iron-rich foods—produces growth trajectories and choking rates comparable to traditional spoon-feeding of purees.

EVIDENCE AND CAUSAL ASSESSMENT

- LEAP RCT (dutoit-2015-leap, N=640 infants aged 4–11 months with severe eczema and/or egg allergy): Infants randomly assigned to consume at least 6 grams of peanut protein per week distributed across three or more meals had a 1.9% prevalence of peanut allergy at 60 months in the skin-prick-negative cohort, compared with 13.7% in the avoidance group (an 86% relative reduction, $p < 0.001$). Across the combined intention-to-treat cohorts, peanut allergy dropped from 17.2% to 3.2% (an 81% relative reduction).

- BLISS RCT (fangupo-2016-bliss, N=206 infants randomized to modified baby-led introduction to solids vs. usual care): No significant differences were observed in the number of choking episodes at 6, 7, 8, 9, or 12 months between groups. Both groups experienced frequent gagging (a normal protective reflex triggered far forward on the tongue) that declined steeply over the second half of the first year.

EFFECT SIZE AND PERSISTENCE

For early peanut introduction, relative risk reduction is ~81% at age 5 (dutoit-2015-leap), and follow-up persistence trials (LEAP-On) demonstrated that tolerance persisted even after 12 months of subsequent peanut cessation. For feeding format (BLW vs. spoon-feeding), differences in BMI z-scores, iron status, and choking incidence are small to null when parents receive basic safety guidance on food texture and iron content.

BIAS, HETEROGENEITY, AND CONFOUNDING

GI attenuation: robust (<20%) (randomized controlled trials directly assign feeding exposure, removing family-level atopic and socioeconomic confounding).

Observational comparisons of baby-led weaning versus puree feeding are heavily confounded by maternal education, breastfeeding duration, and household income; randomized assignment in BLISS removes this selection bias and shows that feeding philosophy per se is not a major driver of weight or developmental outcomes.

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Does not mean infants should ever be given whole peanuts or thick globs of peanut butter (which are severe mechanical airway hazards); peanut protein must be thinned into warm water, breastmilk, formula, or fruit/vegetable puree. It also does not mean parents must choose strictly between purees and finger foods—combining spoon-feeding and soft finger foods is developmentally sound.

WHAT ACTION IS REASONABLY JUSTIFIED

Introduce thinned peanut protein and well-cooked egg around 4 to 6 months once the infant shows postural and oral readiness for solids (especially in infants with eczema, after consulting a pediatrician if eczema is severe), and allow infants to explore both mashed/pureed textures and soft, squishable finger foods while strictly avoiding hard, round choking hazards (whole nuts, whole grapes, raw carrot coins, popcorn).

VERIFIED SOURCES & DOI IDENTIFIERS

- Du Toit et al. (2015), New England Journal of Medicine, LEAP randomized trial (dutoit-2015-leap)
- Fangupo et al. (2016), Pediatrics, BLISS randomized trial on choking and gagging (fangupo-2016-bliss)